

7.1 Waves: Transverse & Longitudinal

Question Paper

Course	CIEA Level Physics
Section	7. Waves
Topic	7.1 Waves: Transverse & Longitudinal
Difficulty	Medium

Time allowed: 40

Score: /27

Percentage: /100

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Question 1

Circular water waves are produced by vibrating dippers at points P and Q, as illustrated in Fig. 4.1.

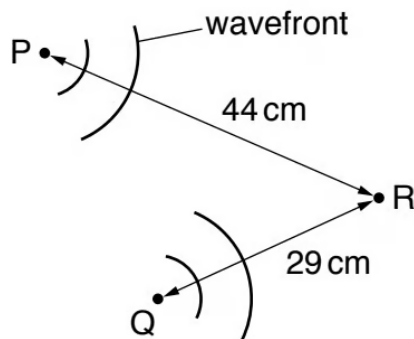


Fig. 4.1 (not to scale)

The waves from P alone have the same amplitude at point R as the waves from Q alone. Distance PR is 44 cm and distance QR is 29 cm.

The dippers vibrate in phase with a period of 1.5 s to produce waves of speed 4.0 cm s^{-1} .

i)

Calculate the wavelength of the waves.

wavelength = cm [2]

ii)

Calculate the path difference at point R of the waves from P and Q.

Give your answer in terms of the wavelength λ of the waves.

path difference = λ [1]

iii)

Describe the motion, if any, of the water particles at point R.

Explain your answer.

[2]

[5 marks]

Question 2a

A wave on the surface of a ripple tank moves from the source at the rear of the tank to the front. The graph in Fig. 1.1 shows the variation with distance x of the displacement y of the surface of the water.

The solid line shows the displacement at $t = 0$ and the dashed line shows the displacement at $t = 0.154$ s.

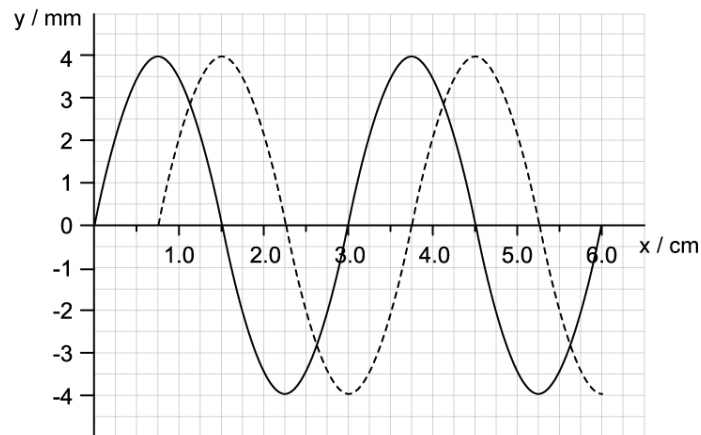


Fig. 1.1

Describe the difference between transverse and longitudinal waves.

[2 marks]

Question 2b

Calculate for the wave on the ripple tank

i)
the speed

[2]

ii)
the frequency

[2]

[4 marks]**Question 2c**

The graph in Fig. 1.1 shows the motion of the water waves at a point where displacement ≤ 6.0 cm.

Describe the appearance of the wave after being displaced by twice this distance.

[3 marks]**Question 2d**

The initial amplitude of the ripples is 0.38 cm.

Sketch a graph of displacement against time to show the motion of the surface of the water for the first 3.0 s.

[3 marks]

Question 3a

Explain what is meant by a progressive wave.

[2 marks]

Question 3b

An elastic cord is fixed at one end and attached to a mechanical oscillator at the other end. Fig. 1.1 shows, at time $t = 0$, the shape of a section of the cord as the wave travels from left to right. W, X, Y and Z are four marked points on the cord.

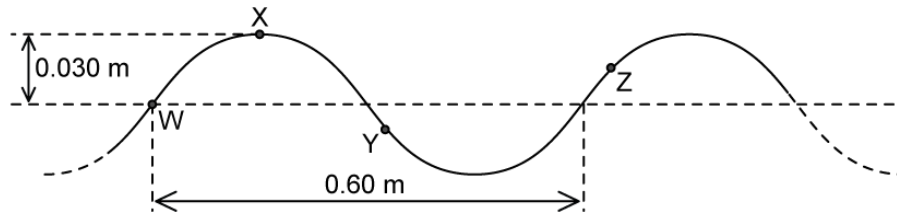


Fig. 1.1

The mechanical oscillator has a steady frequency of 5.0 Hz. The wave has a wavelength of 0.60 m and an amplitude of 0.030 m.

On the axes of Fig. 1.2 sketch the graph of the displacement of point X over the period $t = 0$ to 0.40 s. Add suitable scales to the axes.

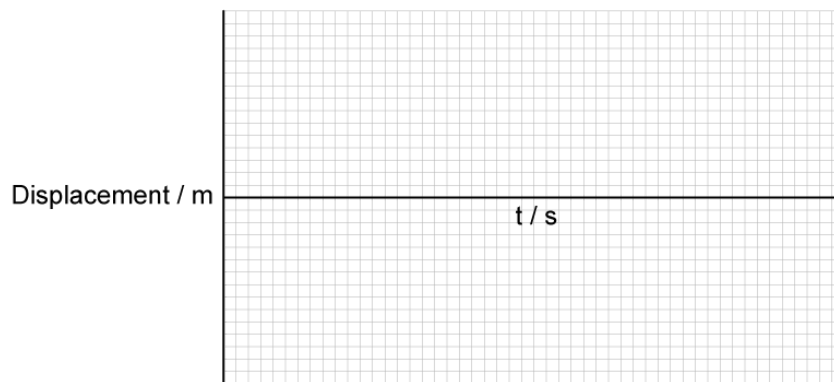


Fig. 1.2

[4 marks]

Question 3c

For time $t = 0$, state which of the points W, X, Y and Z

i)
is instantaneously at rest

[1]

ii)
has the greatest speed

[1]

[2 marks]**Question 3d**

The speed of point W on the cord at $t = 0$ is 0.94 m s^{-1} . With the cord at its original tension, the frequency of oscillation is now doubled to 10 Hz.

The amplitude is kept at 0.030 m. Calculate the new speed of point W at $t = 0$. Explain your reasoning.

[2 marks]

Model Answers: Medium

1

List the known quantities:

- Wave speed, $v = 4.0 \text{ cm s}^{-1}$
- Time period, $T = \frac{1}{f} = 1.5 \text{ s}$
- Distance PR = 44 cm
- Distance QR = 29 cm

i) Use the wave equation to determine wavelength, λ :

- $v = f\lambda$
- $\lambda = \frac{v}{f}$ [1 mark]
- $\lambda = 4.0 \times 1.5 = 6.0 \text{ cm}$ [1 mark]

ii) Calculate the path difference at point R of the waves from P and Q:

- The difference between distances PR and QR = $44 - 29 = 15 \text{ cm}$
- Number of wavelengths in 15 cm = $\frac{15}{6} = \frac{5}{2} = 2.5$
- Path difference = 2.5λ or $\frac{5}{2}\lambda$ [1 mark]

iii) Describe the motion, if any, of the water particles at point R:

- The path difference is an odd number of half wavelengths **OR** the waves have a phase difference of 180° ; [1 mark]
- (This results in destructive interference, hence) the particles are at rest / have no motion; [1 mark]

[Total: 5 marks]

- Spotting that $\lambda = \frac{v}{f} = v \times \frac{1}{f} = v \times T$ can save you some time in part (i)
- Remember to state **path difference** as a **multiple of wavelength** for part (ii)

2a

a) Describe the difference between transverse and longitudinal waves:

- Transverse waves are where the direction of vibration of the particles is perpendicular / at 90° to the direction of motion of the wave / energy transfer; [1 mark]
- Longitudinal waves are where the direction of vibration of the particles are parallel to the direction of motion of the wave / energy transfer; [1 mark]

[Total: 2 marks]

2b

b)

List the known quantities:

- The time between the two positions of the wave, $t = 0.154$ s (from the question in part (a))

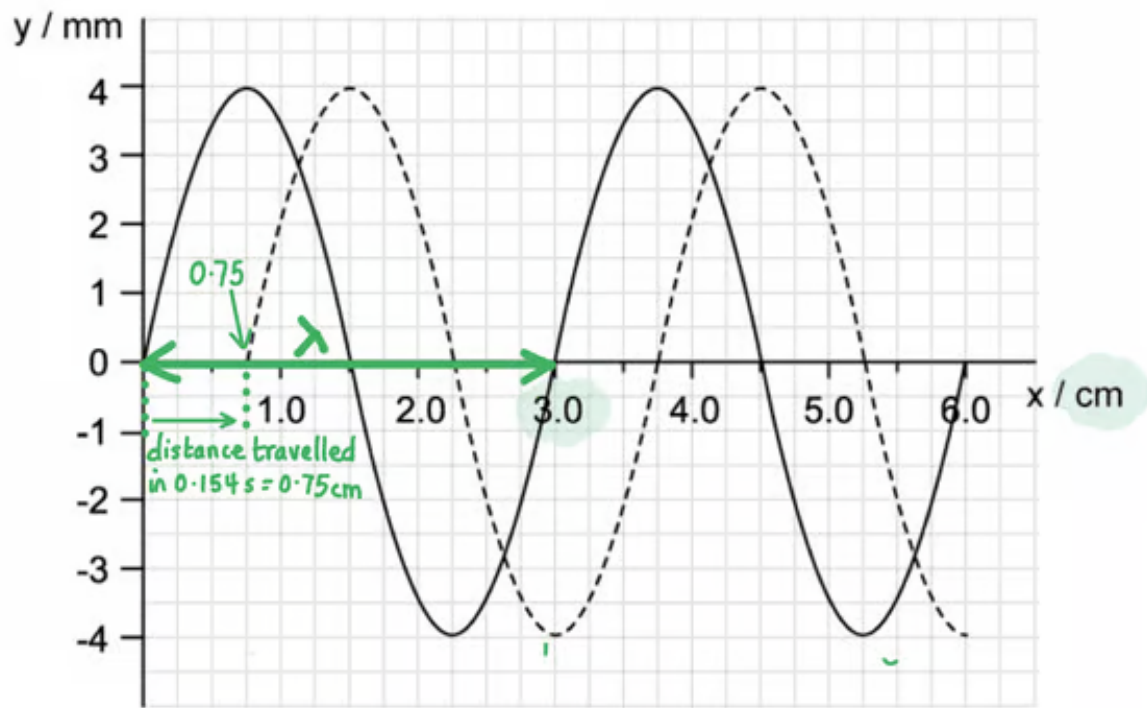
Determine wavelength from the graph, λ :

- $\lambda = 3.0$ cm = 0.03 m

i) Calculating the speed of the wave:

METHOD 1: Calculating using distance and time

From the graph:



- Distance $d = 0.75 \text{ cm} = 7.5 \times 10^{-3} \text{ m}$ [1 mark]

Calculate speed of the wave:

- $v = \frac{d}{t} = \frac{7.5 \times 10^{-3}}{0.154}$
- $v = 0.0487$

Write the answer to two sig figs and include the unit:

- $v = 0.049 \text{ ms}^{-1}$ [1 mark]

METHOD 2: Calculating using frequency and wavelength

From the data booklet:

- $T = \frac{1}{f}$
- $c = f \lambda$

Determine wavelength from the graph, λ :

- $\lambda = 3.0 \text{ cm} = 0.03 \text{ m}$

Calculate time period, T :

- From the graph we can see that the wave progressed by $\frac{1}{4}$ of a full wavelength
- Therefore:
 - $T = 4t$ as the wave progressed by 0.75 cm in that time
- $T = 4t = 0.616 \text{ s}$ [1 mark]

Rearrange time period equation for frequency f and substitute in values:

- $f = \frac{1}{T} = \frac{1}{0.616}$
- $f = 1.62 \text{ Hz}$

Substitute values into wave speed equation:

- $c = 1.62 \times 0.03 = 0.0486$
- $c = 0.049 \text{ m s}^{-1}$ [1 mark]

ii) To find the frequency of the wave:

METHOD 1: calculating using wavelength and speed

List the known quantities:

- Wave speed, $v = 0.049 \text{ m s}^{-1}$ (from part i)

From the data booklet:

- $c = f\lambda$

Calculate f from wave equation:

- $f = \frac{v}{\lambda} = \frac{0.049}{0.03}$ [1 mark]
- $f = 1.6 \text{ Hz}$ [1 mark]

METHOD 2: calculating using time period

From the data booklet:

- $T = \frac{1}{f}$

Calculate time period, T :

- From the graph we can see that the wave progressed by $\frac{1}{4}$ of a full wavelength
- Therefore:

- $c = f \lambda$

Determine wavelength from the graph, λ :

- $\lambda = 3.0 \text{ cm} = 0.03 \text{ m}$

Calculate time period, T :

- From the graph we can see that the wave progressed by $\frac{1}{4}$ of a full wavelength
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- $T = 4t = 0.616 \text{ s}$ [1 mark]

Rearrange time period equation for frequency f and substitute in values:

- $f = \frac{1}{T} = \frac{1}{0.616}$
- $f = 1.62 \text{ Hz}$

Substitute values into wave speed equation:

- $c = 1.62 \times 0.03 = 0.0486$
- $c = 0.049 \text{ m s}^{-1}$ [1 mark]

ii) To find the frequency of the wave:

METHOD 1: calculating using wavelength and speed

List the known quantities:

- Wave speed, $v = 0.049 \text{ m s}^{-1}$ (from part i)

From the data booklet:

- $c = f\lambda$

Calculate f from wave equation:

- $f = \frac{v}{\lambda} = \frac{0.049}{0.03}$ [1 mark]
- $f = 1.6 \text{ Hz}$ [1 mark]

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Calculate time period, T :

- From the graph we can see that the wave progressed by $\frac{1}{4}$ of a full wavelength
- Therefore:
 - $T = 4t$ as the wave progressed by 0.75 cm in that time
- $T = 4t = 0.616 \text{ s}$ [1 mark]

Rearrange time period equation for frequency f and substitute in values:

- $f = \frac{1}{T} = \frac{1}{0.616}$
- $f = 1.6 \text{ Hz}$ [1 mark]

[Total: 4 marks]

2c

c) The appearance of the wave after being displaced twice this distance is:

Any **three** from:

- Amplitude decreases as energy is dissipated; [1 mark]
- Frequency remains the same as it is caused by the source; [1 mark]
- Wavelength remains the same as it is inversely proportional to frequency; [1 mark]
- Speed remains the same as it is proportional to frequency; [1 mark]

[Total: 3 marks]

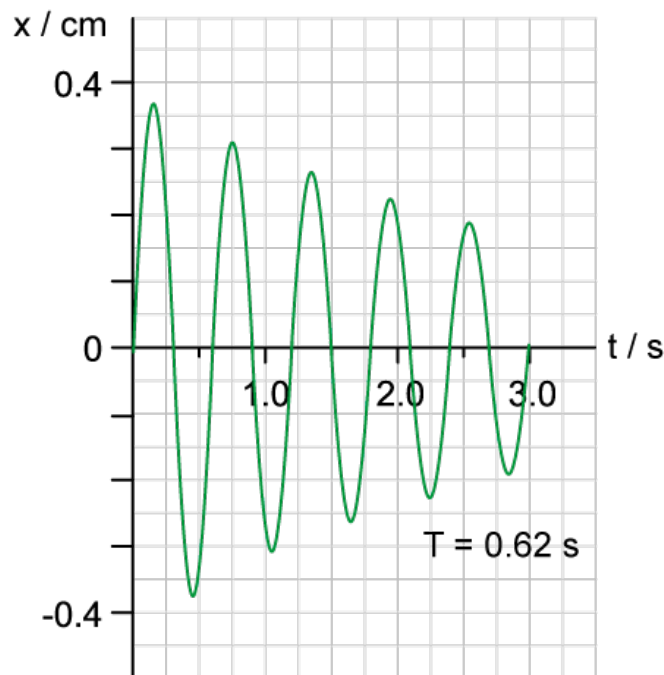
Notice in this question you are asked about the observed **appearance** of the wave. Observations are a key skill in science. Don't confuse those things which you can see (such as amplitude) with others which you can infer mathematically, such as time period.

2d

d) Sketch a graph of displacement against time to show the motion of the surface of

the water for the first 3.0 s:

A graph that would score **three marks** should look like:



- The time period is shown and 5 full waves drawn; [1 mark]
- Time of 3.0 s marked on axes; [1 mark]
- Amplitude gradually decreases as time passes; [1 mark]

[Total: 3 marks]

3a

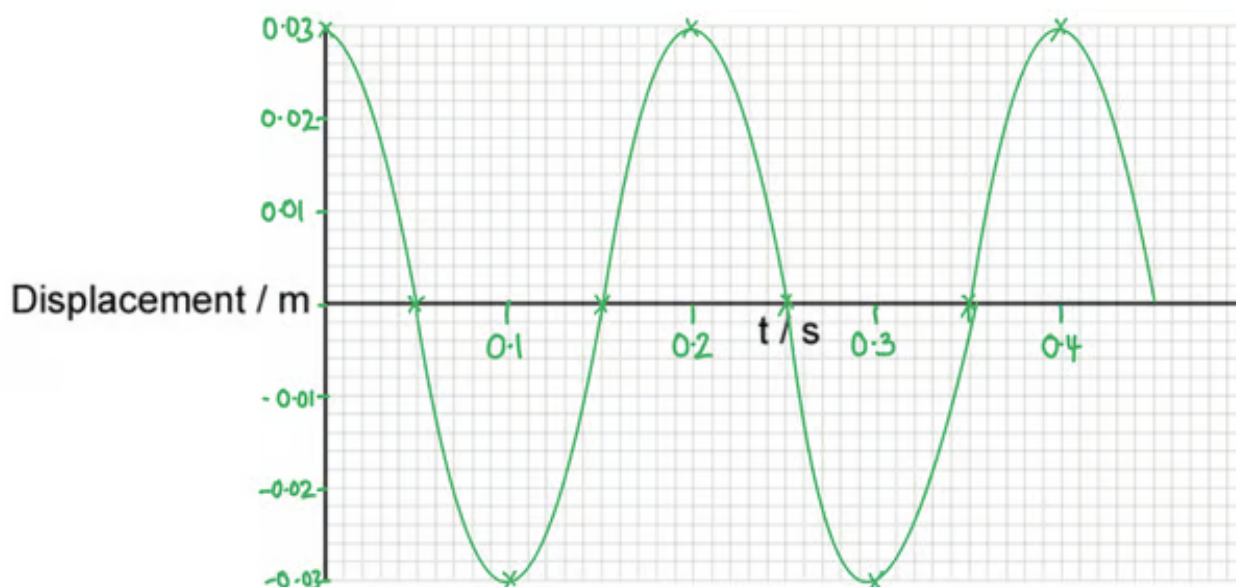
a) Explain what is meant by a progressive wave:

- A transfer of energy; [1 mark]
- With no transfer of matter; [1 mark]

[Total: 2 marks]

3b

b) A correctly completed graph of the oscillation looks like:



- Shape is sinusoidal with exactly two wavelengths shown; [1 mark]
- Amplitude at ± 0.03 ; [1 mark]
- Period = 0.2 s **AND** x-axis labelled 0.1, 0.2, 0.3 and 0.4; [1 mark]
- Curve starts at (0, 0.03); [1 mark]

[Total: 4 marks]

3c

c) For time $t = 0$, state which of the points W, X, Y and Z:

i) Is instantaneously at rest:

- X; [1 mark]

ii) Has the greatest speed:

- W; [1 mark]

[Total: 2 marks]

3d

d) Calculate the new speed of point W at $t = 0$. Explain your reasoning:

- Frequency has doubled so point W has got to move double the distance in the same time **OR** move the same distance in half the time; [1 mark]

- Therefore speed doubles OR new speed = 1.9 m s^{-1} [1 mark]

[Total: 2 marks]

